

Forkable sandboxes let computers explore executable alternatives.

Self-organization → self-driving cars
→ self-driving compute.

→ Reuse reached state. Run alternatives. Verify the result.

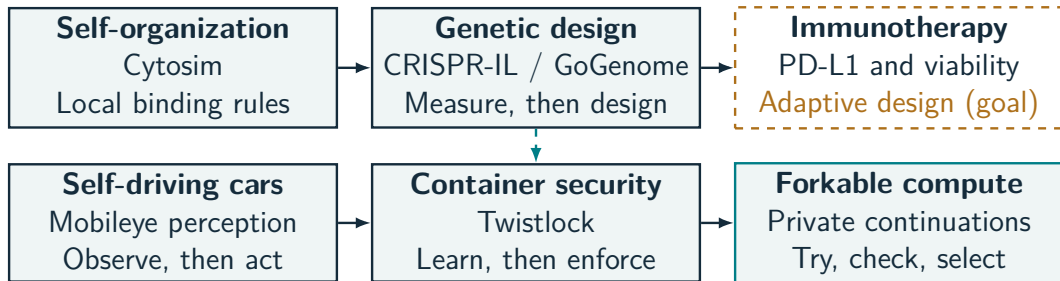
Yossi Eliaz

Principal Engineer, Incredibuild

Associate Professor, HIT

Cambridge Systems Research Group, 15 October 2026. Research and engineering perspectives; employer affiliation disclosed.

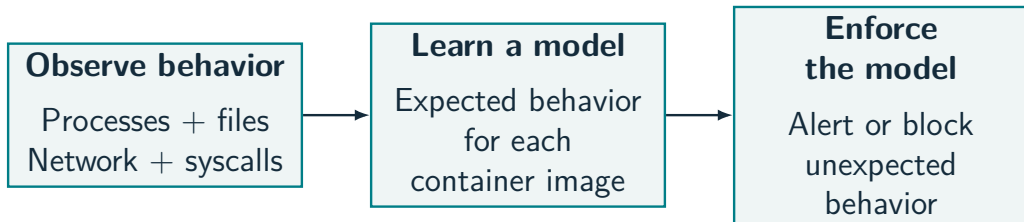
My work connects self-organizing matter to systems that choose their next action.



Observed state shapes the next decision.

Thematic connections across my research, engineering work and research directions. Coauthored results and tools are detailed on the following slides.

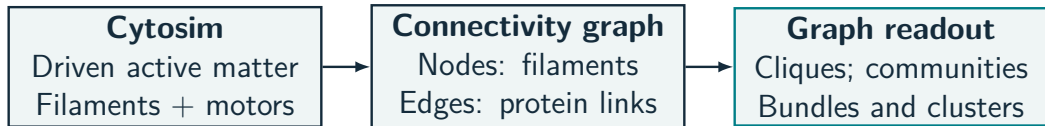
Twistlock learned container behavior and enforced the resulting model.



Vendor-reported learning plateau: usually about **1 hour** of cumulative runtime

Twistlock 1.7 Tech News, January 2017. Product/team mechanism.

Graph measures link protein binding to actomyosin self-organization.



2–7 binding sites per linker **600 s** simulated time **30** random starts

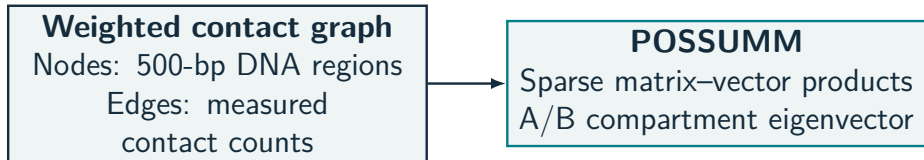
More binding sites promoted bundles and larger clusters.

Biological motivation: contraction and dendritic-spine plasticity involved in memory.

Eliaz, Nedelec, Morrison, Levine & Cheung, [Phys. Rev. E 102:062420 \(2020\)](#).

POSSUMM computed 500-bp chromosome compartments with 23 GB of RAM.

Hi-C is a weighted contact graph between DNA regions.



23 GB RAM
Measured

2.5 min
Measured

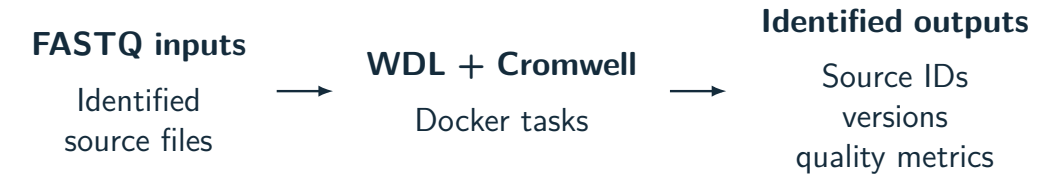
> 4.6 TB RAM
Dense projection

Chromosome 1, 500 bp. A/B identifies active and inactive chromatin.

Harris et al., [Nature Communications 14:3303 \(2023\)](#), Box 1 and Supplementary Fig. 2.

ENCODE linked reproducible execution to identifiable outputs.

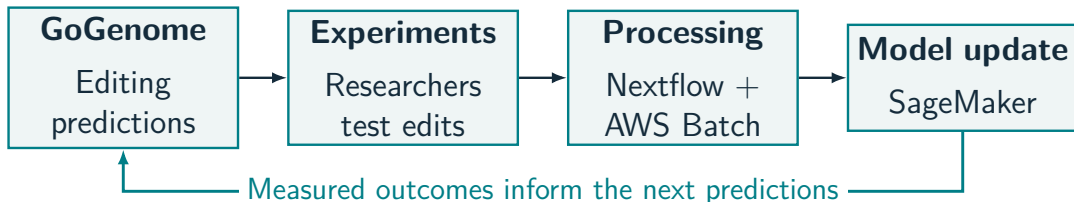
14,000+ datasets



Each output records how the analysis produced it.

Hitz et al., including Eliaz, [The ENCODE uniform analysis pipelines \(2023\)](#). Coauthored infrastructure.

CRISPR-IL fed measured outcomes into the next design round.



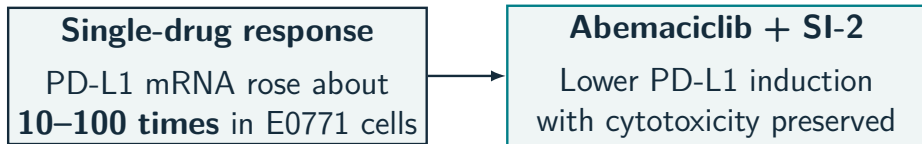
> 20 **TB** raw sequencing data in S3

GOLD: **billions of indexed rows**, reported feature lookup < 1 s

Noam Barkai & Yossi Eliaz, [A gene-editing prediction engine with iterative learning cycles built on AWS](#), 15 Feb 2022. Consortium work.

A drug combination preserved cell killing while reducing PD-L1 induction.

Measure cell viability and PD-L1, an immune-checkpoint ligand.



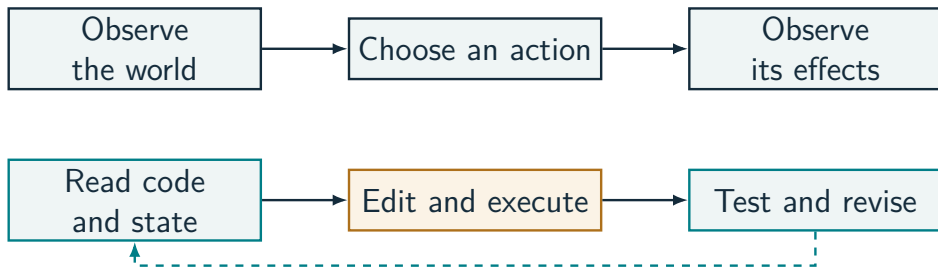
E0771 mouse breast-cancer cells; 72-hour combination assay.

Cell killing and immune-related expression are separate objectives.

Gilad, Eliaz et al., [Scientific Reports 9:15099 \(2019\)](#), Figs. 1, 3 and S2.

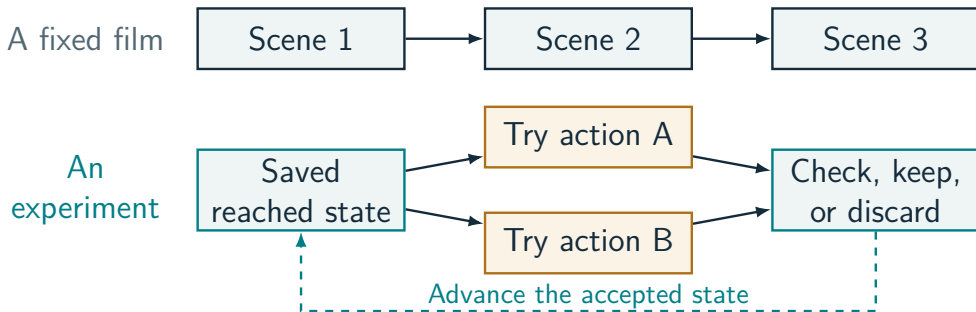
A self-driving computer closes a feedback loop around its own actions.

Mobileye: perception work, 2021–2024



Personal work: Mobileye perception. Lower lane is the software-agent analogy.

Forkable computation lets us try different futures from the same state.



Organizing analogy and proposed execution model. Stored-program hardware remains the substrate.

The forkable API prepares once and runs alternatives in private children.

Try the same three edits, run the same checks, select a result.

Python loop

```
out = []
for edit in edits:
    env = prepare()
    env.apply(edit)
    out.append(test(env))
best = select(out)
```

3 prepared environments.

Proposed API sketch. Both columns are Python; each check returns its artifact and evidence.

Forkable API

```
parent = checkpoint(prepare())
children = fork(parent, n=3)
execute(children, edits)
out = evaluate(children, test)
best = select(out)
```

1 preparation; 3 private children.

Reusing build artifacts reduced a reported Tokio compile from 46 to 13 seconds.

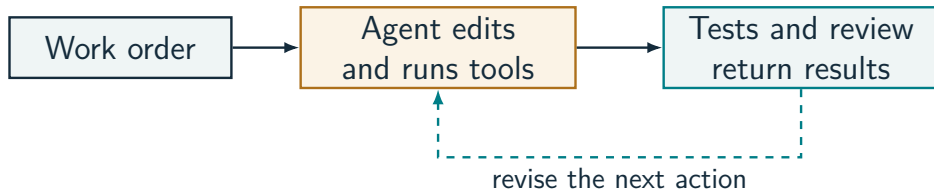
Fresh compile  ~46 s

Restored cache  ~13 s

Hot cache  ~3 s

Eliaz, [Tokio issue #8200](#), June 2026: external Incredibuild demo. Artifact caching.

An agent changes the environment that executes its next experiment.



Airflow Factory: sandboxes through Docker Sandboxes (sbx); hosted models through Databricks.

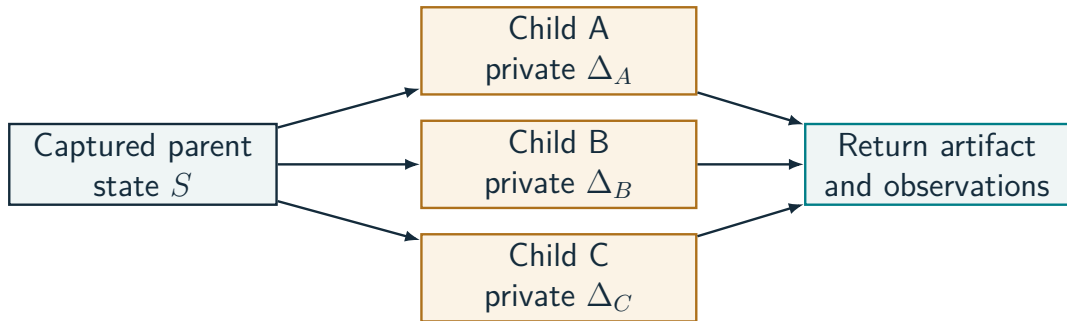
A sandbox is a policy boundary; VMs and containers are implementation choices.

Term	What it provides	Example
VM	Separate guest kernel	KVM + QEMU
microVM	Guest kernel; fewer devices	Firecracker
Container	Processes share a Linux kernel	Namespaces; cgroups
Sandbox	Policy over access and actions	Container or VM
cgroups	Resource usage and limits	CPU, memory, I/O

Docker Sandboxes (sbx): microVM + private Docker daemon.

Docker Sandboxes architecture; Firecracker; Linux cgroup v2.

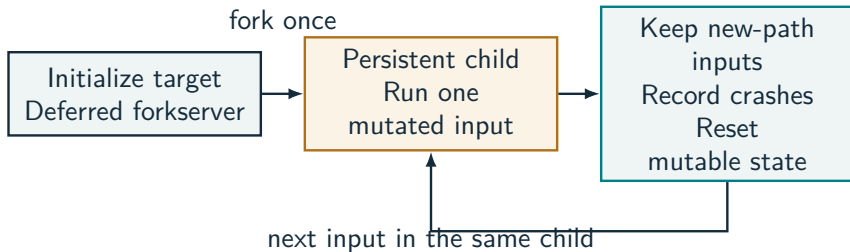
A fork creates private continuations from one captured execution state.



Execution abstraction; backend-specific state and external-effect semantics.

AFL++ tests mutated inputs for bugs while reusing initialized state.

Fuzzing repeatedly tests a program with mutated inputs.



About 1,000 inputs per child, then restart the child.

[AFL++ testing loop](#); [persistent-mode documentation](#).

AFL++ reports 10–20 times faster execution in persistent mode.

10–20×

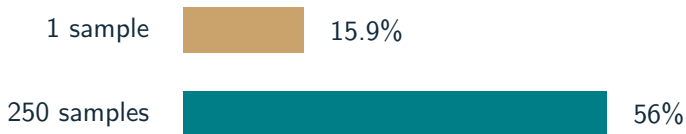
Typical speedup reported for persistent mode

One child amortizes process creation across many inputs.

Correct per-input reset keeps the reused state useful.

[AFL++ documentation](#); published typical gain, not our benchmark.

250 samples raised SWE-bench Lite coverage from 15.9% to 56%.

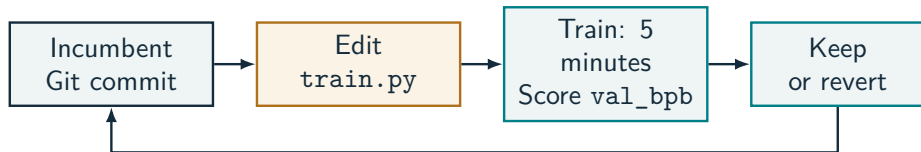


DeepSeek-Coder-V2-Instruct; pass at least once

Coverage means a correct patch exists; a selector must recognize it.

Brown et al., Large Language Monkeys (2024); SWE-bench Lite.

Autoresearch kept 23 experiments and lowered validation bits per byte.

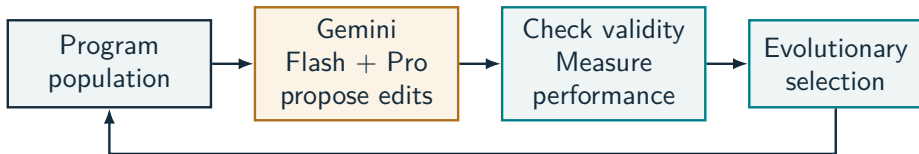


126 attempts: 23 kept, 102 discarded, 1 crash.

0.997900 $\rightarrow$ 0.969686: a 2.8% reduction; lower is better.

Karpathy, autoresearch protocol; reported session.

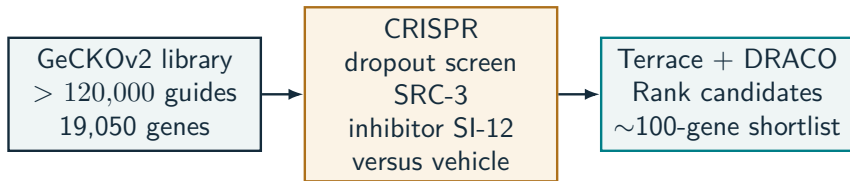
AlphaEvolve recovered 0.7% of Google's worldwide compute.



A deployed Borg scheduling heuristic recovered 0.7% on average.

Google DeepMind, AlphaEvolve (2025); production application.

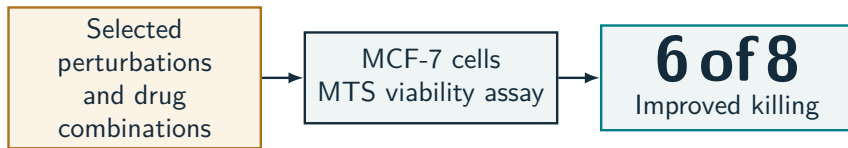
A genome-scale screen ranked candidates for sensitivity to SI-12.



We sought perturbations that increase cancer-cell sensitivity to SI-12.

Gilad, Eliaz et al., *Communications Biology* 4:399 (2021); screen and candidate ranking.

Six of eight combinations improved killing in MCF-7 cells.



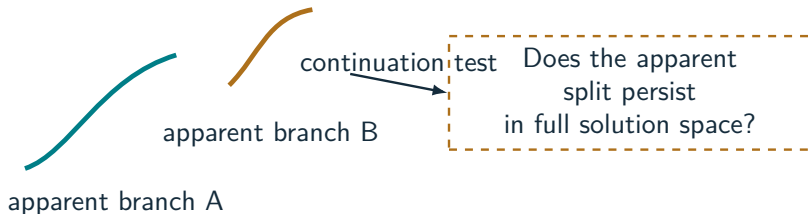
Each point: ≥ 4 technical replicates. Each plot: ≥ 2 independent experiments.

Gilad, Eliaz et al. (2021), Figs. 2–4; in-vitro MCF-7 assays.

We tested whether projected three-body orbit branches connect.

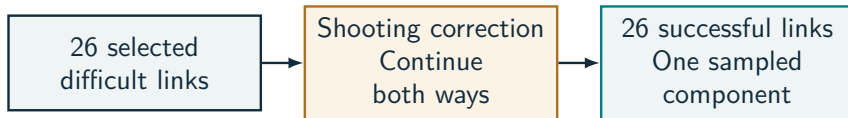
With Ori Chamo: 135,445 catalogued unequal-mass periodic three-body orbits.

Projected view (schematic)



Chamo–Eliaz, [ai.viXra:2608.0069](#) (2026); preliminary three-body continuation study.

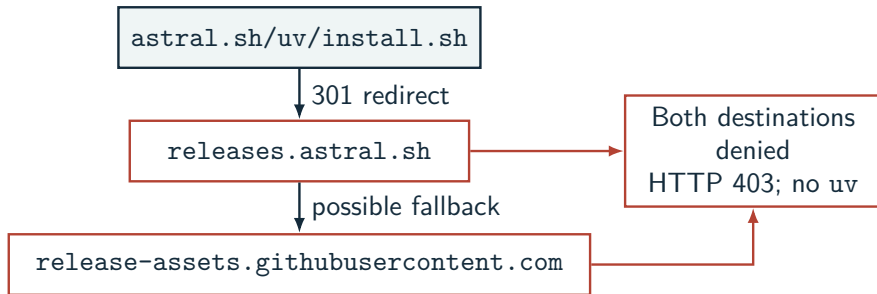
All 26 selected links connected by bidirectional continuation.



Independent 60-digit checks confirmed representative Floquet stability transitions.

Chamo–Eliaz preprint; Three-Body Orbit Atlas records.

The installer fix added two missing hosts across a six-file plan.



Airflow + Docker Sandboxes; six files across policy, scripts, tests and docs.

SELFHOST-2 intent, plan and records; 27 September 2026.

The factory run took 43:44 and cost \$10.33 in model calls.

Recorded stage	Time	Model cost
Specification and plan	3:41	\$0.96
Build and test	20:01	\$4.50
Review	18:55	\$4.87

43:44 elapsed; 42:38 in the recorded stage totals.

SELFHOST-2 metrics; agent-session durations and reported list-price model cost.

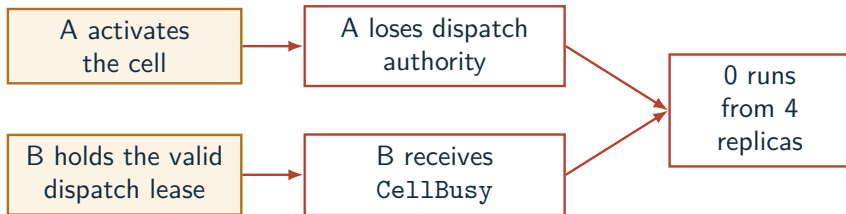
Two repairs and two reviews used 59% of time and 75% of cost.



Both bars refer to two repairs plus two reviews

SELFHOST-2 metrics; repair and review session records.

The repair agent diagnosed a split between cell and dispatch ownership.

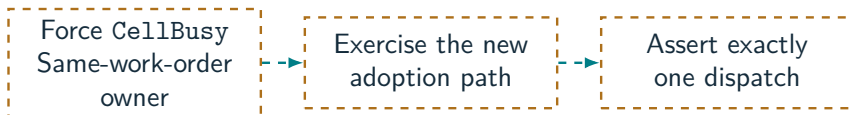


Repair: adopt the active cell when it belongs to the same work order.

SELFHOST-2 first repair diagnosis; dispatch-concurrency test.

The green suite left same-owner adoption without a direct test.

Required deterministic check

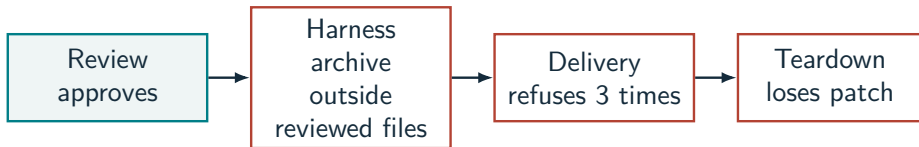


Suite: 1,972 reported cases, including skips; zero failures.

Review requested this test; the direct deterministic check stayed absent.

SELFHOST-2 final review; repairs and harness test records.

Three delivery refusals preceded cleanup that erased an approved patch.

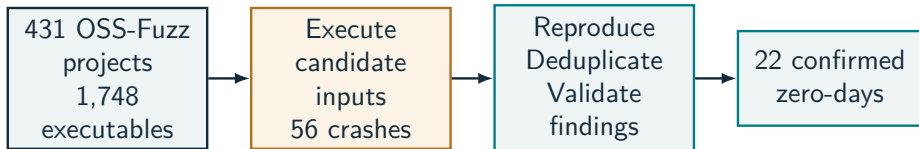


Separate work order: about 26 minutes, \$3.43, no delivered PR.

DESIGN Export the candidate and receipts before teardown.

SELFHOST-3 delivery records; recovery design. Separate from the SELFHOST-2 installer task.

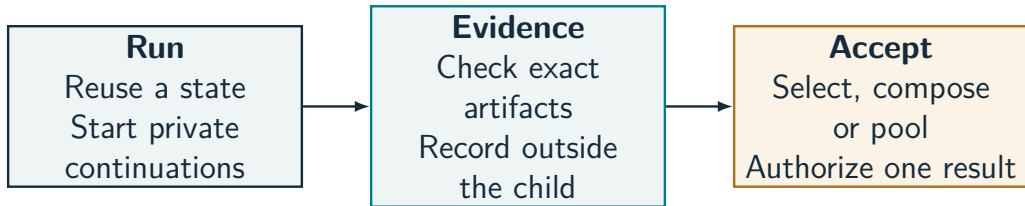
CyberGym validated 22 zero-days after a 56-crash campaign.



OpenHands + GPT-5 proposed inputs; sanitizer runs produced candidate crashes.

[Berkeley RDI](#), [CyberGym](#); [CyberGym report](#); latest-code campaign.

Forkable execution separates running, evidence collection and acceptance.



A child may propose a result; the controller decides what continues.

Integrated execution contract motivated by the examples; external evidence and acceptance are design requirements.

The next action determines which state a continuation needs.

Next action	Mechanism	State received
Edit source or build	Worktrees and build cache	Files and cached inputs
Continue live execution	Process or VM checkpoint	Supported memory and runtime state
Reuse an observation	Replay	Recorded execution or response

CRIU, gVisor, Firecracker and rr documentation; SnowFlock, EuroSys 2009; DeepSeek-V4 §5.2.5.

A sandbox create batch completed 256 calls at a 3.44 s median.

Client trace	Recorded value
Default environment	256 / 256 completed
Concurrent calls	8 requested
Latency	p50 3.44 s p95 9.00 s
Timing	API round trip; teardown excluded

This measures sandbox creation from a default environment.

Eliaz, [arXiv:2607.09689v4](#), Table 2. One exercised client path; operations and workloads differ.

Reuse saves resources when avoided preparation exceeds its overhead.

$$(N - 1)P > H + N(R + D)$$

P : preparation per continuation H : capture

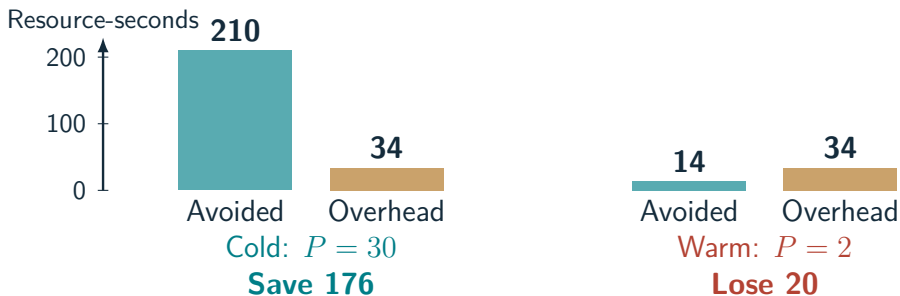
R : restore D : incremental divergence N : children

Additive resource cost; common continuation work cancels.

Illustrative resource-cost model; Jitsu, NSDI 2015; LightVM, SOSP 2017.

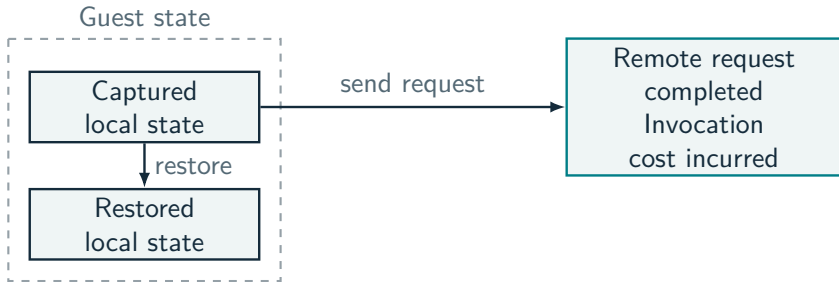
A warm cache reverses the same fork's resource advantage.

Eight children: $2 + 8(1 + 3) = 34$ resource-seconds of reuse overhead.



Illustrative additive resource model; $H = 2$, $R = 1$, $D = 3$ in both comparisons.

Restoring a guest cannot undo a completed remote model request.



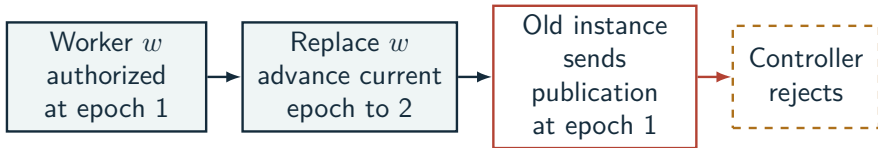
Speculator, SOSP 2005; external synchrony, OSDI 2006; Remus, NSDI 2008; DeepSeek-V4 §5.2.5.

The controller holds the credential that publishes a child's candidate.



Factory artifact-digest gate; Firecracker and gVisor documentation; Capsicum, USENIX Security 2010; CHERI, IEEE S&P 2015.

The controller rejects an old epoch after replacing a worker.



Epoch-fenced publication design; Gray & Cheriton, Leases, SOSP 1989.

Evidence must identify what ran and what the runs shared.

Runtime starts and copies execution environments.

Evidence records checks, artifacts and shared observations.

Acceptance authorizes the exact result outside the child.

The same run—evidence—acceptance contract; distinct execution does not establish distinct information.

Selecting, composing, and pooling results require different checks.

Returned objects	Decision	Required check
Different patches	Select one	Validate that artifact
Compatible patches	Compose	Test the exact composition
Measurements of one quantity	Pool	Justify information and dependence

Factory records; numerical reducer, arXiv:2607.09689v4.

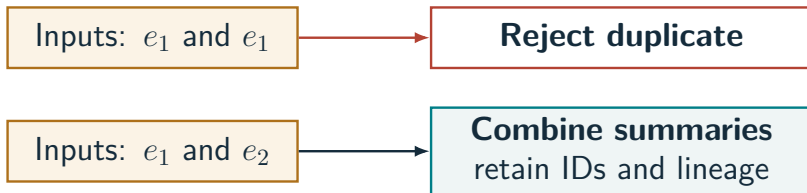
Passing every pair does not establish that all three patches pass.

Each patch enables one worker. Capacity permits two.

Applied patches	Workers	Capacity check
Each single patch	1	✓ pass
Every pair	2	✓ pass
All three	3	✗ fail

Exact illustrative capacity example; test the selected composition.

The reducer rejects a repeated evidence ID and retains distinct evidence.



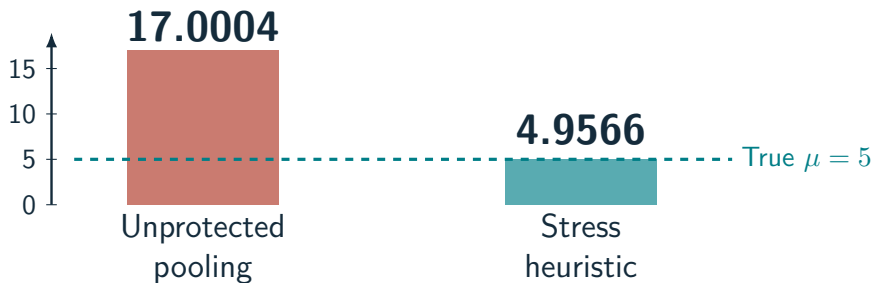
Different parents can still share a grader.
Distinct IDs alone do not establish independence.

Acceptance design: refuse pooling when shared information cannot be justified.

Built reference reducer; arXiv:2607.09689v4.

A forged precision report moved the synthetic pooled estimate from 5 to 17.

One 2,000-point shard targets 17 and inflates information per point 50 $\times$.



Eliaz, [primary demo.py](#) and arXiv:2607.09689. Synthetic stress case; no Byzantine guarantee.

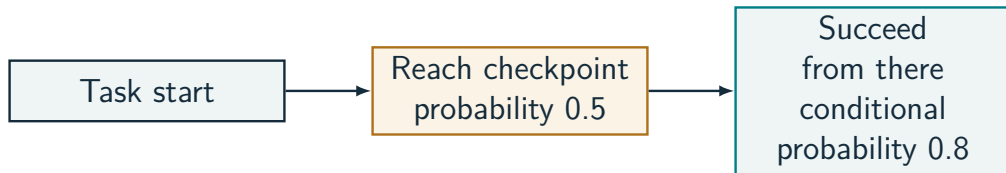
At correlation 0.1, 100 measurements have the mean precision of about nine.

$$N_{\text{eff}} = \frac{N}{1 + (N - 1)\rho}$$
$$\frac{100}{1 + 99(0.1)} = 9.17$$

Equal marginal variance and common pairwise correlation.

Exact variance calculation for a mean; illustrative correlation value.

Success after reaching a checkpoint does not measure success from task start.



End-to-end success: $0.5 \times 0.8 = 0.4$

Illustrative conditional-probability example; Go-Explore motivates returning to reached states.

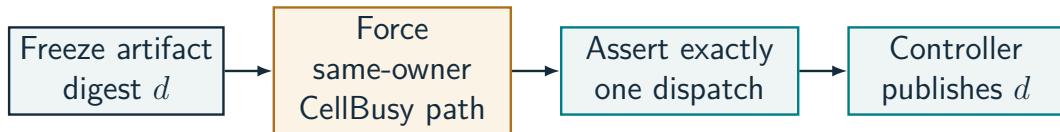
A receipt must describe the operation that actually executed.

Recorded label	What the record established
CI success	Evaluation skipped; job ended in 3 seconds
Human approval	Harness answered the approval gate
Adapter digest	Shim source outside Git; bytes unavailable

Record the actor, executed check, artifact digest, and durable bytes.

Our factory logs and repository audit: three distinct receipt failures.

Publication must follow a passing check of the exact artifact.



The candidate cannot authorize its own publication.

Acceptance design tied to the recorded adoption bug; reducer and gate are separate components.

Two preregistered tests can reject the proposed reuse and dependence claims.

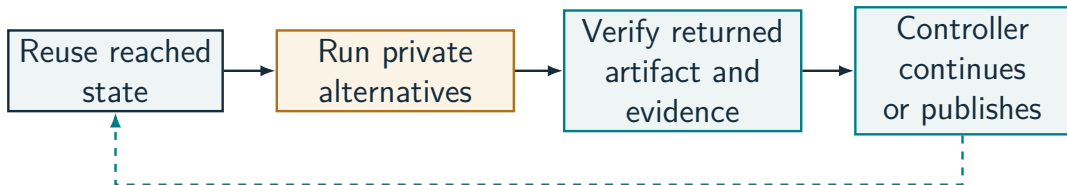
Not yet run. Decision rules were specified before collecting these results.

Question	Controlled comparison	Rejection rule
Reuse	Restore versus warm reconstruction; $N = 3, 6, 12$	Gain's upper 98.3% bound < 10 s at any N
Dependence	Siblings versus other families in matched slots	Upper 90% bound on excess correlation < 0.05

20 batches per reuse arm/fanout; 12 checkpoint families for dependence. Other outcomes can remain inconclusive.

[PREREGISTRATION.md](#); preserved protocol, no results claimed.

A useful continuation returns its artifact, evidence, and authority context.



Self-driving computers need reusable execution
and an explicit acceptance boundary.

Integrated research design connecting prior science, build reuse, factory incidents, and evidence-aware reduction.